Assignment No - 1

1 Unit 1: First order Ordinary D.E.

Type A: Exact Differential Equations

Q1) Solve $(2xy^4 + \sin y)\,dx + (4x^2y^3 + x\cos y)\,dy = 0$.

→ Given $(2xy^4 + \sin y)\,dx + (4x^2y^3 + x\cos y)\,dy = 0$

compare given D.E with $M\,dx + N\,dy = 0$

$M = 2xy^4 + \sin y$ $\qquad$ $N = 4x^2y^3 + x\cos y$.

$\frac{\partial M}{\partial y} = 8xy^3 + \cos y$ $\qquad$ $\frac{\partial N}{\partial x} = 8xy^3 + \cos y$.

x const $\qquad$ y const.

Therefore $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

Hence give D.E is exact D.E.

⇒ Sol^n for exact D.E

$$\underset{y\ \text{const}}{\int M\,dx} + \int \left[\begin{array}{c}\text{terms of N does} \\ \text{not containing } x\end{array}\right] dy = C$$

$$\Rightarrow \int (2xy^4 + \sin y)\,dx + \int (0)\,dy = C$$

$$\Rightarrow 2y^4 \int x\,dx + \sin y \int 1\,dx + 0 = C$$

$$\Rightarrow 2y^4 \left[\frac{x^2}{2}\right] + \sin y\,[x] = C$$

$$\boxed{x^2y^4 + x\sin y = C}$$ ✓

Q2) Solve $(3y^2 + 10xy^2)\,dx + (6xy - 2 + 10x^2y)\,dy = 0$

$\longrightarrow$ Given $(3y^2 + 10xy^2)\,dx + (6xy - 2 + 10x^2y)\,dy = 0$

compare given D.E with $M\,dx + N\,dy = 0$

$M = 3y^2 + 10xy^2$ $\qquad$ $N = 6xy - 2 + 10x^2y$

$\underset{x \text{ const}}{\frac{\partial M}{\partial y}} = 6y + 20xy$ $\qquad$ $\underset{y \text{ const}}{\frac{\partial N}{\partial x}} = 6y + 20xy$

Therefore $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$\therefore$ Hence given D.E is exact D.E.

$\Rightarrow$ Soln for D.E.

$$\underset{y = \text{const}}{\int M\,dx} + \int \left[\begin{array}{l}\text{terms of N does} \\ \text{not contain } x\end{array}\right] dy = C$$

$$\underset{y = \text{const}}{\int (3y^2 + 10xy^2)\,dx} + \int (-2)\,dy = C$$

$$3y^2 \underbrace{\int (1)\,dx}_{x} + 10y^2 \underbrace{\int x\,dx}_{\frac{x^2}{2}} + \underbrace{\int (-2)\,dy}_{-2y} = C$$

$$\boxed{3xy^2 + 5y^2x^2 - 2y = C}$$

Q3) Solve $(y\sec^2 x + \sec x \cdot \tan x)\,dx + (\tan x + 2y)\,dy = 0$

→ Given $(y\sec^2 x + \sec x \cdot \tan x)\,dx + (\tan x + 2y)\,dy = 0$

compare given DE with $M\,dx + N\,dy = 0$.

$M = y\sec^2 x + \sec x \tan x$ $\qquad$ $N = \tan x + 2y$

$\underset{x\ \text{const}}{\frac{\partial M}{\partial y}} = \sec^2 x$ $\qquad$ $\underset{y = \text{const}}{\frac{\partial N}{\partial x}} = \sec^2 x$

There fore $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

∴ Hence given DE is exact D.E.

⇒ Sol^n for D.E

$$\underset{y\ \text{const}}{\int M\,dx} + \int \left[\begin{array}{c}\text{terms of N does} \\ \text{not contain } x\end{array}\right] dy = C$$

$$\underset{y = \text{const}}{\int (y\sec^2 x + \sec x \tan x)\,dx} + \int (2y)\,dy = C$$

$$y\int \sec^2 x\,dx + \int \sec x \tan x\,dx + \int (2y)\,dy = C$$

$$\boxed{y\tan x + \sec x + y^2 = C}$$ ✓

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Q4) Solve $(x^2y - 2xy^2)dx + (3x^2y - x^3)dy = 0$

→ Given $(x^2y - 2xy^2)dx + (3x^2y - x^3)dy = 0 \longrightarrow$ (I)

Compare given eqn with $Mdx + Ndy = 0$

$M = x^2y - 2xy^2$ $\qquad$ $N = 3x^2y - x^3$

$\frac{\partial M}{\partial y} = x^2 - 4xy$ $\qquad$ $\frac{\partial N}{\partial x} = 6xy - 3x^2$

x const. $\qquad$ y const.

Therefore $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

∴ Hence give DE is non exact DE

⟹ Observed that M and N are Homogeneous function of same degree 3.

Therefore $IF = \frac{1}{Mx + Ny}$

$$IF = \frac{1}{x(x^2y - 2xy^2) + y(3x^2y - x^3)}$$

$$IF = \frac{1}{\cancel{x^3y} - 2x^2y^2 + 3x^2y^2 - \cancel{x^3y}}$$

$$\boxed{IF = \frac{1}{x^2y^2}}$$

⟹ multiply IF to the eqn (I)

$$\frac{1}{x^2y^2}(x^2y - 2xy^2)dx + \frac{1}{x^2y^2}(3x^2y - x^3)dy = 0$$

$$\left(\frac{x^2y}{x^2y^2} - \frac{2xy^2}{x^2y^2}\right)dx + \left(\frac{3x^2y}{x^2y^2} - \frac{x^3}{x^2y^2}\right)dy = 0$$

$$\left(\frac{1}{y} - \frac{2}{x}\right)dx + \left(\frac{3}{y} - \frac{x}{y^2}\right)dy = 0 \longrightarrow \text{II}$$

$\Rightarrow$ Compare eq^n II with $M_1\,dx + N_1\,dy = 0$

$$M_1 = \frac{1}{y} - \frac{2}{x} \qquad N_1 = \frac{3}{y} - \frac{x}{y^2}$$

$$\underset{x \text{ const}}{\frac{\partial M_1}{\partial y}} = -\frac{1}{y^2} \qquad \underset{y \text{ const}}{\frac{\partial N_1}{\partial x}} = -\frac{1}{y^2}$$

Therefore $\dfrac{\partial M_1}{\partial y} = \dfrac{\partial N_1}{\partial x}$

Hence eq^n II is exact D.E.

$\Rightarrow$ Sol^n will be

$$\underset{y \text{ const}}{\int M_1\,dx} + \int \left[\begin{array}{c}\text{terms of } N_1 \text{ does} \\ \text{not containing } x\end{array}\right] dy = C$$

$$\int\left(\frac{1}{y} - \frac{2}{x}\right)dx + \int\left(\frac{3}{y}\right)dy = C$$

$$\frac{1}{y}\int(1)\,dx - 2\int\left(\frac{1}{x}\right)dx + 3\int\left(\frac{1}{y}\right)dy = C$$

$$\boxed{\frac{x}{y} - 2\log x + 3\log y = C} \quad (\text{OR}) \quad \boxed{\frac{x}{y} + \log\left(\frac{y^3}{x^2}\right) = C}$$

..... $\log a - \log b = \log\left(\frac{a}{b}\right)$

Q5) Solve $(xy + y^2)\,dx - x^2 dy = 0$.

$\longrightarrow$ Given $(xy + y^2)\,dx - x^2 dy = 0 \longrightarrow$ (I)

Compare given eqn with $M dx + N dy = 0$.

$M = xy + y^2$ $\qquad$ $N = -x^2$

$\underset{x \text{ const}}{\frac{\partial M}{\partial y}} = x + 2y$ $\qquad$ $\underset{y \text{ const}}{\frac{\partial N}{\partial x}} = -2x$

Therefore $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

Hence given eqn is not exact (non-exact) D.E.

$\Rightarrow$ Observed that M and N are Homogeneous function of some degree 2.

Therefore $IF = \frac{1}{Mx + Ny}$

$$IF = \frac{1}{x(xy + y^2) + y(-x^2)}$$

$$IF = \frac{1}{\cancel{x^2y} + xy^2 - \cancel{x^2y}}$$

$$\boxed{IF = \frac{1}{xy^2}}$$

$\Rightarrow$ Multiply I.F. to the eq^n (I)

$$\frac{1}{xy^2}(xy + y^2)\,dx + \frac{1}{xy^2}(-x^2)\,dy = 0$$

$$\left(\frac{xy}{xy^2} + \frac{y^2}{x}\right)dx - \left(\frac{x^2}{xy^2}\right)dy = 0$$

$$\left(\frac{1}{y} + \frac{1}{x}\right)dx - \left(\frac{x}{y^2}\right)dy = 0 \longrightarrow \text{(II)}$$

$\Rightarrow$ Compare eq^n (II) with $M_1\,dx + N_1\,dy = 0$.

$$M_1 = \frac{1}{y} + \frac{1}{x} \qquad N_1 = -\frac{x}{y^2}$$

$$\underset{x\ \text{const}}{\frac{\partial M_1}{\partial y}} = -\frac{1}{y^2} \qquad \underset{y\ \text{const}}{\frac{\partial N_1}{\partial x}} = -\frac{1}{y^2}$$

Therefore $\dfrac{\partial M_1}{\partial y} = \dfrac{\partial N_1}{\partial x}$

Hence eq^n (II) is exact D.E.

$\Rightarrow$ Sol^n of D.E.

$$\int_{y=\text{const}} M_1\,dx + \int \left[\begin{array}{l}\text{terms of } N_1 \text{ does} \\ \text{not containing } x\end{array}\right] dy = C$$

$$\int_{y=\text{const}} \left(\frac{1}{y} + \frac{1}{x}\right) dx + \int (0)\,dy = C$$

$$\frac{1}{y}\int (1)\,dx + \int \left(\frac{1}{x}\right) dx = c$$

$$\boxed{\frac{x}{y} + \log x = c}$$ ✓

Q6) Solve $(x^2y^2 + 5xy + 2)y\,dx + (x^2y^2 + 4xy + 2)x\,dy = 0$

→ Given $(x^2y^2 + 5xy + 2)y\,dx + (x^2y^2 + 4xy + 2)x\,dy = 0 \rightarrow (I)$

Compare given eqn with $M\,dx + N\,dy = 0$

$M = (x^2y^2 + 5xy + 2)y$ $\qquad$ $N = (x^2y^2 + 4xy + 2)x$

$M = x^2y^3 + 5xy^2 + 2y$ $\qquad$ $N = x^3y^2 + 4x^2y + 2x$

$\frac{\partial M}{\partial y} = 3x^2y^2 + 10xy + 2$ $\qquad$ $\frac{\partial N}{\partial x} = 3x^2y^2 + 8xy + 2$

x const $\qquad$ y const

Therefore $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

∴ Hence give D.E is non-exact D.E

⟹ observed that M and N are ~~Homogenous function of same~~ in the form of

$$y f_1(xy)\,dx + x f_2(xy)\,dy = 0$$

$$I.F. = \frac{1}{Mx - Ny} = \frac{1}{x(x^2y^3 + 5xy^2 + 2y) - y(x^3y^2 + 4x^2y + 2x)}$$

$$I.F. = \frac{1}{\cancel{x^3y^3} + 5x^2y^2 + \cancel{2xy} - \cancel{x^3y^3} - 4x^2y^2 - \cancel{2xy}}$$

$$\boxed{I.F = \frac{1}{x^2y^2}}$$

$\Rightarrow$ Multiply I.F. to the eq^n (I).

$$\frac{1}{x^2y^2}(x^2y^3 + 5xy^2 + 2y)\,dx + \frac{1}{x^2y^2}(x^3y^2 + 4x^2y + 2x)\,dy = 0.$$

$$\left(\frac{x^2y^3}{x^2y^2} + \frac{5xy^2}{x^2y^2} + \frac{2y}{x^2y^2}\right)dx + \left(\frac{x^3y^2}{x^2y^2} + \frac{4x^2y}{x^2y^2} + \frac{2x}{x^2y^2}\right)dy = 0.$$

$$\left(y + \frac{5}{x} + \frac{2}{x^2y}\right)dx + \left(x + \frac{4}{y} + \frac{2}{xy^2}\right)dy = 0 \quad \text{--- (II)}$$

$\Rightarrow$ Compare with $M_1\,dx + N_1\,dy = 0$.

$$M_1 = \left(y + \frac{5}{x} + \frac{2}{x^2y}\right) \qquad N_1 = \left(x + \frac{4}{y} + \frac{2}{xy^2}\right)$$

$$\underset{x\ \text{const}}{\frac{\partial M_1}{\partial y}} = 1 - \frac{2}{x^2y^2} \qquad \underset{y\ \text{const}}{\frac{\partial N_1}{\partial x}} = 1 - \frac{2}{x^2y^2}$$

Therefore $\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$

$\therefore$ Hence eq^n (II) is exact D.E.

$\Rightarrow$ Sol^n of D.E.

$$\int_{y\,\text{const}} M\,dx + \int \left[\begin{array}{l}\text{terms of } N \text{ does} \\ \text{not containing } x\end{array}\right] dy = C$$

$$\int \left(y + \frac{5}{x} + \frac{2}{x^2 y}\right) dx + \int \left(\frac{4}{y}\right) dy = C$$

$$y\int (1)\,dx + 5\int \left(\frac{1}{x}\right) dx + \frac{2}{y}\int \left(\frac{1}{x^2}\right) dx + 4\int \left(\frac{1}{y}\right) dy = C$$

$$-\frac{1}{x}$$

$$xy + 5\log x - \frac{2}{xy} + 4\log y = C$$

$$\boxed{xy + 5\log x + 4\log y - \frac{2}{xy} = C}$$ ✓

Q7) Solve $(x^2y^2+2)y\,dx + (2-2x^2y^2)x\,dy = 0$.

$\rightarrow$ Given $(x^2y^2+2)y\,dx + (2-2x^2y^2)x\,dy = 0 \longrightarrow$ (I)

Compare given eq^n with $M\,dx + N\,dy = 0$.

$M = (x^2y^2+2)y$ $\qquad$ $N = (2-2x^2y^2)x$

$M = x^2y^3 + 2y$ $\qquad$ $N = 2x - 2x^3y^2$

$\underset{x\,\text{const}}{\frac{\partial M}{\partial y}} = 3x^2y^2 + 2$ $\qquad$ $\underset{y\,\text{const}}{\frac{\partial N}{\partial x}} = 2 - 6x^2y^2$

Therefore $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

$\therefore$ Hence given eq^n is non-exact.

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→ Given DE is in the form of $y f_1(xy)\,dx + x f_2(xy)\,dy = 0$

$$I.F = \frac{1}{Mx - Ny} = \frac{1}{x(x^2y^3 + 2y) - y(2x - 2x^3y^2)}$$

$$I.F = \frac{1}{x^3y^3 + \cancel{2xy} - \cancel{2xy} + 2x^3y^3}$$

$$\boxed{I.F = \frac{1}{3x^3y^3}}$$ ✓

→ Multiply IF to the eqn ①

$$\frac{1}{3x^3y^3}(x^2y^2 + 2)y\,dx + \frac{1}{3x^3y^3}(2 - 2x^2y^2)x\,dy = 0.$$

$$\left(\frac{x^2y^3 + 2y}{3x^3y^3}\right)dx + \left(\frac{2x - 2x^3y^2}{3x^3y^3}\right)dy = 0$$

$$\left(\frac{x^2y^3}{3x^3y^3} + \frac{2y}{3x^3y^3}\right)dx + \left(\frac{2x}{3x^3y^3} - \frac{2x^3y^2}{3x^3y^3}\right)dy = 0$$

Multiply by 3 both the sides we get.

$$\left(\frac{1}{x} + \frac{2}{x^3y^2}\right)dx + \left(\frac{2}{x^2y^3} - \frac{2}{y}\right)dy = 0.$$

$\Rightarrow$ Compare with $M_1 dx + N_1 dy = 0$

$$M_1 = \frac{1}{x} + \frac{2}{x^3y^2} \qquad N_1 = \frac{2}{x^2y^3} - \frac{2}{y}$$

$$\underset{x \text{ const}}{\frac{\partial M_1}{\partial y}} = \frac{2}{x^3} \times \left(-\frac{2}{y^3}\right) \qquad \underset{y \text{ const}}{\frac{\partial N_1}{\partial x}} = \frac{2}{y^3} \times \left(-\frac{2}{x^3}\right)$$

Therefore $\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$

$\therefore$ Hence given eq^n (II) is exact D.E

$\Rightarrow$ Sol^n of D.E.

$$\underset{y\text{const}}{\int M_1 dx} + \int \left[\begin{array}{c}\text{terms of } N_1 \text{ does} \\ \text{not containing } x\end{array}\right] dy = C$$

$$\underset{y\text{const}}{\int} \left(\frac{1}{x} + \frac{2}{x^3y^2}\right) dx + \int \left(-\frac{2}{y}\right) dy = C$$

$$\int \left(\frac{1}{x}\right) dx + \int \left(\frac{2}{x^3y^2}\right) dx - 2\int \left(\frac{1}{y}\right) dy = C$$

$$\log x - \frac{1}{x^2y^2} - 2\log y = C \quad \ldots \left\{\begin{array}{l} n\log x = \log x^n \\ \int\left(\frac{1}{x^3}\right) = -\frac{1}{2x^2}\end{array}\right\}$$

$$\boxed{\log\left(\frac{x}{y^2}\right) - \frac{1}{x^2y^2} = C}$$

Q8) Solve $(x^2+y^2+1)\,dx - 2xy\,dy = 0$

$\rightarrow$ Given $(x^2+y^2+1)\,dx - 2xy\,dy = 0 \longrightarrow$ (I)

Compare with $M\,dx + N\,dy = 0$.

$M = x^2+y^2+1$ $\qquad$ $N = -2xy$

$\frac{\partial M}{\partial y} = 2y$ $\qquad$ $\frac{\partial N}{\partial x} = -2y$.

x const $\qquad$ y const

$\therefore$ Therefore $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

$\therefore$ Hence given eq^n is non-exact D.E.

$\Rightarrow$ Consider,

$$f(x) = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{(2y-(-2y))}{-2xy} = \frac{4y}{-2xy} = -\frac{2}{x}$$

$\longrightarrow$ only f^n of x

$$\Rightarrow IF = e^{\int f(x)\,dx} = e^{\int(-\frac{2}{x})dx} = e^{-2\int(\frac{1}{x})dx}$$

$$IF = e^{-2\log x} \qquad \boxed{IF = x^{-2} = \frac{1}{x^2}}$$

$\Rightarrow$ Multiply IF to eq^n (I).

$$\frac{1}{x^2}(x^2+y^2+1)\,dx - \frac{1}{x^2}(2xy)\,dy = 0.$$

$$\left(\frac{x^2}{x^2} + \frac{y^2}{x^2} + \frac{1}{x^2}\right)dx - \left(\frac{2xy}{x^2}\right)dy = 0.$$

$$\left(1 + \frac{y^2}{x^2} + \frac{1}{x^2}\right)dx - \left(\frac{2y}{x}\right)dy = 0 \longrightarrow \text{(II)}$$

$\Rightarrow$ Compare eqn (II) with $M_1 dx + N_1 dy = 0$

$$M_1 = 1 + \frac{y^2}{x^2} + \frac{1}{x^2} \qquad N_1 = -\frac{2y}{x}$$

$$\underset{x\,\text{const}}{\frac{\partial M_1}{\partial y}} = \frac{2y}{x^2} \qquad \underset{y\,\text{const}}{\frac{\partial N_1}{\partial x}} = \frac{2y}{x^2}$$

Therefore $\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$

Hence eqn (II) is in exact form / O.E.

$\Rightarrow$ Soln of D.E.

$$\underset{y\,\text{const}}{\int M_1\,dx} + \int \left[\begin{array}{c}\text{terms of } N_1 \text{ does} \\ \text{not contain } x\end{array}\right] dy = C$$

$$\underset{y\,\text{const}}{\int}\left(1 + \frac{y^2}{x^2} + \frac{1}{x^2}\right) + \int (0)\,dy = C$$

$$\int (1)\,dx + y^2\int\left(\frac{1}{x^2}\right)dx + \int\left(\frac{1}{x^2}\right)dx = C \qquad \rightarrow -\frac{1}{x}$$

$$\boxed{x - \frac{y^2}{x} - \frac{1}{x} = C}$$

OR

$$\boxed{x^2 - y^2 - 1 = Cx}$$

Q9) Solve $(x^2+y^2+x)\,dx + xy\,dy = 0$.

$\rightarrow$ Given $(x^2+y^2+x)\,dx + xy\,dy = 0 \longrightarrow$ (I)

Compare given eqn with $M\,dx + N\,dy = 0$

$M = x^2+y^2+x$ $\qquad$ $N = xy$

$\underset{x\,\text{const}}{\frac{\partial M}{\partial y}} = 2y$ $\qquad$ $\underset{y\,\text{const}}{\frac{\partial N}{\partial x}} = y$

Therefore $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

$\therefore$ Hence given is non-exact D.E.

$\Rightarrow$ Consider

only f^n of x

$$f(x) = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y-y}{xy} = \frac{y}{xy} = \frac{1}{x}$$

$$IF = e^{\int f(x)\,dx} = e^{\int (\frac{1}{x})\,dx} = e^{\log x} = x$$

$\Rightarrow$ multiply IF to eqn (I)

$x(x^2+y^2+x)\,dx + x(xy)\,dy = 0$

$(x^3+xy^2+x^2)\,dx + x^2y\,dy = 0 \longrightarrow$ (II)

$\Rightarrow$ Compare eqn (II) with $M_1\,dx + N_1\,dy = 0$.

$M_1 = x^3+xy^2+x^2$ $\qquad$ $N_1 = x^2y$

$\underset{x\,\text{const}}{\frac{\partial M_1}{\partial y}} = 2xy$ $\qquad$ $\underset{y\,\text{const}}{\frac{\partial N_1}{\partial x}} = 2xy$

$\therefore$ Therefore $\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$

$\therefore$ Hence given eqn (II) are exact D.E.

$\Rightarrow$ Soln of D.E.

$$\underset{y \text{ const}}{\int M_1\,dx} + \int \left[\begin{array}{c}\text{terms of N does} \\ \text{not containing } x\end{array}\right] dy = C$$

$$\int (x^3 + xy^2 + x^2)\,dx + \int (0)\,dy = C$$

$$\int (x^3)\,dx + y^2 \int (x)\,dx + \int (x^2)\,dx = C$$

$$\boxed{\frac{x^4}{4} + \frac{x^2}{2}y^2 + \frac{x^3}{3} = C} \quad \ldots \text{(multiply by 12 both the sides)}$$

OR

$$\boxed{3x^4 + 6x^2y^2 + 4x^3 = C_1} \checkmark$$

Q10) Solve $(2x^2y^2 + ye^x)dx + (-e^x - y^3)dy = 0$

$\longrightarrow$ Given $(2x^2y^2 + ye^x)\,dx + (-e^x - y^3)\,dy = 0 \longrightarrow$ (I)

Compare with $M\,dx + N\,dy = 0$

$M = 2x^2y^2 + ye^x$ $\qquad$ $N = -e^x - y^3$

$\underset{x \text{ const}}{\frac{\partial M}{\partial y}} = 4x^2y + e^x$ $\qquad$ $\underset{y \text{ const}}{\frac{\partial N}{\partial x}} = -e^x$

Therefore $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

Hence

$\therefore$ given eqn is non-exact D.E.

$\Rightarrow$ Consider,

$$g(y) = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{(-e^x - 4x^2y - e^x)}{2x^2y^2 + ye^x}$$

$$g(y) = \frac{(-2e^x - 4x^2y)}{2x^2y^2 + ye^x} = -\frac{2}{y}\left(\frac{\cancel{e^x + 2x^2y}}{\cancel{2x^2y + e^x}}\right)$$

$$\boxed{g(y) = -\frac{2}{y}}$$ only function of y.

$$I.F. = e^{\int g(y)\,dy} = e^{\int(-\frac{2}{y})dy} = e^{-2\log y}$$

$$IF = e^{\log y^{-2}} = y^{-2} = \frac{1}{y^2}$$

$\Rightarrow$ Multiply IF to the eqn (I).

$$\frac{1}{y^2}(2x^2y^2 + ye^x)\,dx + \frac{1}{y^2}(-e^x - y^3)\,dy = 0.$$

$$\left(\frac{2x^2y^2}{y^2} + \frac{ye^x}{y^2}\right)dx + \left(\frac{-e^x}{y^2} - \frac{y^3}{y^2}\right)dy = 0$$

$$\left(2x^2 + \frac{e^x}{y}\right)dx + \left(\frac{-e^x}{y^2} - y\right)dy = 0 \longrightarrow \text{(II)}$$

Compare with $M_1\,dx + N_1\,dy = 0$

$M_1 = 2x^2 + \frac{e^x}{y}$ $\qquad$ $N_1 = -\frac{e^x}{y^2} - y.$

$$M_1 = 2x^2 + \frac{e^x}{y} \qquad N_1 = -\frac{e^x}{y^2} - y$$

$$\frac{\partial M_1}{\partial y} = -\frac{e^x}{y^2} \qquad \frac{\partial N_1}{\partial x} = -\frac{e^x}{y^2}$$

x const y const

Therefore $\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$

$\therefore$ Hence eq^n (II) is exact D. E.

$\Rightarrow$ Sol^n of D. E.

$$\underset{y \text{ const}}{\int M_1\,dx} + \int \left[\begin{array}{l}\text{terms of } N_1 \text{ does} \\ \text{not contains } x\end{array}\right] dy = C.$$

$$\underset{y \text{ const}}{\int \left(2x^2 + \frac{e^x}{y}\right) dx} + \int (-y)\,dy = C$$

$$2\int (x^2)\,dx + \frac{1}{y}\int (e^x)\,dx + -1\int (y)\,dy = C$$

$$\boxed{\frac{2x^3}{3} + \frac{e^x}{y} - \frac{y^2}{2} = C}$$

Q11) Solve $(y^4 + 2y)\,dx + (xy^3 + 2y^4 - 4x)\,dy = 0$

$\rightarrow$ Given $(y^4 + 2y)\,dx + (xy^3 + 2y^4 - 4x)\,dy = 0 \longrightarrow$ (I)

Compare with $M\,dx + N\,dy = 0$

$M = y^4 + 2y$ $\qquad\qquad$ $N = xy^3 + 2y^4 - 4x$

$\frac{\partial M}{\partial y} = 4y^3 + 2$ $\qquad\qquad$ $\frac{\partial N}{\partial x} = y^3 - 4$

x const $\qquad\qquad$ y const

Therefore $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

$\therefore$ Hence given eqn is non-exact D.E.

$\Rightarrow$ Consider,

$$g(y) = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{(y^3 - 4) - (4y^3 + 2)}{y^4 + 2y}$$

only fn of y

$$g(y) = \frac{-3y^3 - 6}{y(y^3 + 2)} = \frac{-3}{y}\,\frac{(y^3 + 2)}{(y^3 + 2)} = -\frac{3}{y}$$

$$I.F. = e^{\int g(y)\,dy} = e^{\int(-\frac{3}{y})\,dy} = e^{-3\int(\frac{1}{y})\,dy}$$

$$IF = e^{-3\log y} = e^{\log y^{-3}} = y^{-3} = \frac{1}{y^3}$$

$\Rightarrow$ Multiply I.F. to the eqn (I).

$$\frac{1}{y^3}(y^4 + 2y)\,dy + \frac{1}{y^3}(xy^3 + 2y^4 - 4x)\,dy = 0$$

$$\left(\frac{y^4}{y^3} + \frac{2y}{y^3}\right) dy + \left(\frac{xy^3}{y^3} + \frac{2y^4}{y^3} - \frac{4x}{y^3}\right) dx = 0$$

$$\left(y + \frac{2}{y^2}\right) dy + \left(x + 2y - \frac{4x}{y^3}\right) dx = 0 \longrightarrow \text{(II)}$$

eq^n (II)

$\Rightarrow$ Compare^n with $M_1 dx + N_1 dy = 0$.

$M_1 = y + \frac{2}{y^2} \rightarrow \left(\partial_{y^2} = -\frac{2}{y^3}\right)$ $\qquad N_1 = x + 2y - \frac{4x}{y^3}$

$\frac{\partial M_1}{\partial y} = 1 + \frac{4}{y^3}$ $\qquad \frac{\partial N_1}{\partial x} = 1 - \frac{4}{y^3}$

x const $\qquad$ y const

Therefore $\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$.

$\therefore$ Hence eq^n (II) is exact D.E.

$\Rightarrow$ Sol^n of D.E.

$$\int_{y \text{ const}} M_1 dx + \int \left[\begin{array}{l}\text{terms of } N_1 \text{ does} \\ \text{not contains } x\end{array}\right] dy = C.$$

$$\int_{y=\text{const}} \left(y + \frac{2}{y^2}\right) dx + \int (2y)\, dy = C.$$

$$y\int(1)dx + \frac{2}{y^2}\int(1)dx + 2\int(y)dy = C$$

$$\boxed{xy + \frac{2x}{y^2} + y^2 = C}$$ ✓

Q12) Solve $(y^3 - 2x^2y)dx + (2xy^2 - x^3)dy = 0$.

→ Given $(y^3 - 2x^2y)dx + (2xy^2 - x^3)dy = 0 \longrightarrow (I)$

Compare given eqn with $Mdx + Ndy = 0$.

$M = y^3 - 2x^2y$ $\qquad$ $N = 2xy^2 - x^3$

$\underset{x\,\text{const}}{\frac{\partial M}{\partial y}} = 3y^2 - 2x^2$ $\qquad$ $\underset{y\,\text{const}}{\frac{\partial N}{\partial x}} = 2y^2 - 3x^2$

Therefore $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

∴ Hence given eqn is non-exact D.E.

⇒ observed that M and N are Homogeneous function of same degree 3

$$I.F = \frac{1}{Mx + Ny} = \frac{1}{x(y^3 - 2x^2y) + y(2xy^2 - x^3)}$$

$$IF = \frac{1}{\cancel{xy^3} - 2x^3y + 2xy^3 - \cancel{x^3y}}$$

$$\boxed{IF = \frac{1}{3xy(y^2 - x^2)}}$$

⇒ Multiply I.F to the eqn (I).

$$\frac{1}{3xy(y^2 - x^2)}(y^3 - 2x^2y)\,dx + \frac{1}{3xy(y^2 - x^2)}(2xy^2 - x^3)\,dy = 0$$

(multiply by)

$$\frac{y^3 - 2x^2y}{3xy(y^2 - x^2)}\,dx + \frac{2xy^2 - x^3}{3xy(y^2 - x^2)}\,dy = 0.$$

$$\frac{y(y^2 - 2x^2)}{xy(y^2 - x^2)}\,dx + \frac{x(2y^2 - x^2)}{xy(y^2 - x^2)}\,dy = 0$$

$$\frac{y^2 - 2x^2}{x(y^2 - x^2)}\,dx + \frac{2y^2 - x^2}{y(y^2 - x^2)}\,dy = 0$$

$$\frac{(y^2 - x^2) - x^2}{x(y^2 - x^2)}\,dx + \frac{(y^2 - x^2) + y^2}{y(y^2 - x^2)}\,dy = 0$$

$$\left(\frac{1}{x} - \frac{x}{y^2 - x^2}\right)dx + \left(\frac{1}{y} + \frac{y}{y^2 - x^2}\right)dy = 0 \quad \text{--- (II)}$$

$\Rightarrow$ Compare eqn (II) with $M_1\,dx + N_1\,dy = 0$.

$$M_1 = \frac{1}{x} - \frac{x}{y^2 - x^2} \qquad N_1 = \frac{1}{y} + \frac{y}{y^2 - x^2}$$

$$\underbrace{\frac{\partial M_1}{\partial y}}_{x \text{ const}} = \frac{\partial}{\partial y}\left(\frac{x}{y^2 - x^2}\right) \qquad \underbrace{\frac{\partial N_1}{\partial x}}_{y \text{ const}} = \frac{\partial}{\partial x}\left(\frac{y}{y^2 - x^2}\right)$$

$$\frac{\partial M_1}{\partial y} = \frac{2xy}{(y^2 - x^2)^2} \qquad \frac{\partial N_1}{\partial x} = \frac{2xy}{(y^2 - x^2)^2}$$

Therefore $\dfrac{\partial M_1}{\partial y} = \dfrac{\partial N_1}{\partial x}$

$\therefore$ Hence given eqn is exact D.E.

$\Rightarrow$ Sol^n of D.E.

$$\int_{y\ \text{const}} M_1\,dx + \int \left[\begin{array}{l}\text{terms of } N_1 \text{ does} \\ \text{not contain } x\end{array}\right] dy = C$$

$$\int \left(\frac{1}{x} - \frac{x}{y^2 - x^2} \right) dx + \int \left(\frac{1}{y}\right) dy = C.$$

$$\int \left(\frac{1}{x}\right) dx + \int \left(\frac{x}{y^2 - x^2} \right) dx + \int \left(\frac{1}{y}\right) dy = C.$$

$$\log x + \frac{1}{2} \log (y^2 - x^2) + \log y = C.$$

$$\log (xy) + \log \sqrt{y^2 - x^2} = C$$

$$\log \left(xy \sqrt{y^2 - x^2} \right) = C \quad \text{..... (Exponential (e))}.$$

$$xy \sqrt{y^2 - x^2} = c' \quad \text{..... (Squaring)}$$

$$x^2 y^2 (y^2 - x^2) = c''$$

$$\boxed{x^2 y^4 - x^4 y^2 = c_1}$$

* Type C: Linear Differential Equations *

Q13) Solve $x^2 \frac{dy}{dx} = 3x^2 - 2xy + 1$

→ Given $x^2 \frac{dy}{dx} = 3x^2 - 2xy + 1$

$\Rightarrow \frac{dy}{dx} = 3 - \frac{2y}{x} + \frac{1}{x^2}$ ..... (Divided by x^2)

$\Rightarrow \frac{dy}{dx} + \left(\frac{2}{x}\right) y = 3 + \frac{1}{x^2} \longrightarrow (*)$

⇒ This is first order Linear Differential Equation (LDE). Compare with $\frac{dy}{dx} + P(x)\, y = Q(x)$

i.e $P(x) = \frac{2}{x}$, $Q(x) = 3 + \frac{1}{x^2}$

Integrating factor :-

If $I = e^{\int P(x)dx} = e^{\int \frac{2}{x} dx} = e^{2 \log x} = e^{\log x^2}$

$\boxed{IF = I = x^2}$

⇒ Sol^n will be

$y \cdot I = \int I \cdot Q\, dx + C$

$\Rightarrow y \cdot x^2 = \int x^2 \left(3 + \frac{1}{x^2}\right) dx + C$

$\Rightarrow yx^2 = \int (3x^2 + 1)\, dx + C$

$$\Rightarrow yx^2 = \cancel{3}\left[\frac{x^3}{\cancel{3}}\right] + x + c$$

$$\Rightarrow \boxed{yx^2 = x^3 + x + c}$$

OR

$$\Rightarrow \boxed{y = x + \frac{1}{x} + \frac{c}{x^2}}$$

Q14) Solve $(x^2+1)\frac{dy}{dx} + 4xy = \frac{1}{(x^2+1)^2}$

$\longrightarrow$ Given $(x^2+1)\frac{dy}{dx} + 4xy = \frac{1}{(x^2+1)^2}$

... divide by (x^2+1)

$$\Rightarrow \frac{dy}{dx} + \left(\frac{4x}{(x^2+1)}\right)y = \frac{1}{(x^2+1)^3} \longrightarrow (*)$$

$\Rightarrow$ This is first order LDE of the form

$\frac{dy}{dx} + Py = Q$ where,

$$P(x) = \frac{4x}{x^2+1}, \quad Q(x) = \frac{1}{(x^2+1)^3}$$

$$IF = I = e^{\int\left(\frac{4x}{x^2+1}\right)dx} = e^{2\int\left(\frac{2x}{x^2+1}\right)dx}$$

$$I = e^{2\log(x^2+1)} = (x^2+1)^2$$

$\boxed{I = (x^2+1)^2} \longrightarrow$ Integrating factor

$\Rightarrow$ Sol^n for first order LDE.

$$y \cdot I = \int I \, Q \, dx + C$$

$$\Rightarrow y(x^2+1)^2 = \int (x^2+1)^2 \cdot \frac{1}{(x^2+1)^3} \, dx + C$$

$$\Rightarrow y(x^2+1)^2 = \int \frac{1}{x^2+1} \, dx + C$$

$$\Rightarrow \boxed{y(x^2+1)^2 = \tan^{-1}(x) + C}$$

OR

$$\boxed{y = \frac{\tan^{-1}(x)}{(x^2+1)^2} + \frac{C}{(x^2+1)^2}}$$

Q15) Solve $\frac{dy}{dx} = \frac{e^x - 3xy}{x^2}$

$\longrightarrow$ Given $\frac{dy}{dx} = \frac{e^x - 3xy}{x^2}$

$$\frac{dy}{dx} = \frac{e^x}{x^2} - \frac{3xy}{x^2}$$

$$\Rightarrow \frac{dy}{dx} + \left(\frac{3}{x}\right) y = \frac{e^x}{x^2} \longrightarrow (*)$$

$\Rightarrow$ This is first order LDE of the form $\frac{dy}{dx} + Py = Q$ where,

$$P(x) = \frac{3}{x}, \quad Q(x) = \frac{e^x}{x^2}$$

$\Rightarrow$ $IF = e^{\int P(x)\,dx} = e^{\int \left(\frac{3}{x}\right) dx} = e^{3\log x} = e^{\log x^3}$

$\boxed{IF = x^3}$

$\Rightarrow$ Sol^n for first order LDE,

$y.I = \int I.Q\,dx + C$

$\Rightarrow y.x^3 = \int x^3 \frac{e^x}{x^2}\,dx + C$

$\Rightarrow y.x^3 = \int x.e^x\,dx + C$

$\Rightarrow y.x^3 = \left[xe^x - \int 1.e^x\right] + C$

$\Rightarrow y.x^3 = xe^x - e^x + C$

$\boxed{y.x^3 = (x-1)e^x + C}$ ✓

Q(6) $x\cos x \frac{dy}{dx} + (\cos x - x\sin x)y = 1$

$\rightarrow$ Given $x\cos x \frac{dy}{dx} + (\cos x - x\sin x)y = 1$

Divide by $x\cos x$

$\frac{dy}{dx} + \left(\frac{1}{x} - \tan x\right)y = \frac{1}{x\cos x}$ ——— $\rightarrow$ (*)

$\Rightarrow$ This is first order LDE of the form

$$\frac{dy}{dx} + Py = Q \quad \text{where}$$

$$P(x) = \frac{1}{x} - \tan x \qquad Q(x) = \frac{\sec x}{x}$$

$$IF = e^{\int P(x)dx} = e^{\int (\frac{1}{x} - \tan x)\, dx}$$

$$IF = e^{\log x - \log \sec x}$$

$$I.F = e^{\log\left(\frac{x}{\sec x}\right)} = \boxed{\frac{x}{\sec x} = I}$$

$\Rightarrow$ Sol^n for first order LDE,

$$y.I = \int I.Q\,dx + C$$

$$\Rightarrow y.\frac{x}{\sec x} = \int \left[\frac{x}{\sec x}\cdot\left(\frac{\sec x}{x}\right)\right] dx + C$$

$$\Rightarrow y.\frac{x}{\sec x} = \int (1)\,dx + C$$

$$\Rightarrow y.\frac{x}{\sec x} = x + C$$

$$\boxed{y\, x\cos x = x + C}$$ ✓

* Type 0: Reducible to L.D.E *

Q17) Solve $xy - \frac{dy}{dx} = y^3 e^{-x^2}$

$\rightarrow$ Given $xy - \frac{dy}{dx} = y^3 e^{-x^2}$

$$\Rightarrow \frac{dy}{dx} - xy = -y^3 e^{-x^2} \longrightarrow \text{(I)}$$

$\Rightarrow$ This is Bernoulli's D.E of the type,

$\frac{dy}{dx} + Py = Q y^n$ with $n = 3$,

where $P = -x$, $Q = -e^{-x^2}$

Divide eq^n (I) by y^3

$$y^{-3}\frac{dy}{dx} - xy^{-2} = -e^{-x^2} \longrightarrow \text{(II)}$$

$\Rightarrow$ Put $y^{1-n} = u$ i.e $\boxed{y^{-2} = u}$

diff wrt x.

$$\boxed{y^{-2} \longrightarrow y \longrightarrow x}$$

$$-2y^{-3}\frac{dy}{dx} = \frac{du}{dx}$$

$$y^{-3}\frac{dy}{dx} = -\frac{1}{2}\frac{du}{dx}$$

eqⁿ (II) becomes

$$-\frac{1}{2}\frac{du}{dx} - xu = -e^{-x^2}$$

$$\Rightarrow \frac{du}{dx} + 2xu = 2e^{-x^2} \quad \text{..... (multiply by -2)}$$

$\Rightarrow$ This is first order LDE of the type $\frac{du}{dx} + pu = Q$ with,

$$P = 2x \qquad Q = 2e^{-x^2}$$

$$IF = I = e^{\int p\,dx} = e^{\int 2x\,dx} = e^{2\left[\frac{x^2}{2}\right]} = e^{x^2}$$

$\Rightarrow$ Solⁿ for first order LDE.

$$u.I = \int I\,Q\,dx + C$$

$$\Rightarrow u.I = \int e^{x^2}\left(2e^{-x^2}\right)dx + C$$

$$\Rightarrow u.I = \int 2e^{(x^2-x^2)}\,dx + C$$

$$\Rightarrow u.e^{x^2} = 2\int(1)\,dx + C$$

$$\Rightarrow u.e^{x^2} = 2x + C$$

Back substitute $u = y^{-2}$

$$\boxed{y^{-2}e^{x^2} = 2x + C} \quad \text{OR} \quad \boxed{\frac{e^{x^2}}{y^2} = 2x + C}$$

Q18) Solve $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)e^x \sec y$.

→ Given $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)e^x \sec y$.

$$\Rightarrow \frac{dy}{dx} - \frac{\sin y}{\cos y\,(1+x)} = (1+x)\,e^x \sec y.$$

multiply by $\cos y$.

$$\cos y \frac{dy}{dx} - \frac{\sin y}{(1+x)} = (1+x)e^x \quad \text{.....} \quad (\sec y \cos y = 1) \quad \text{(I)}$$

→ This D.E of the form $f'(y)\frac{dy}{dx} + f(y)P = Q$.

where P & Q are the functions of x (or) const.

Here $f(y) = \sin y$.

Put $\boxed{\sin y = u}$

$$\cos y \frac{dy}{dx} = \frac{du}{dx}$$

⇒ Eqn (I) becomes

$$\frac{du}{dx} - \frac{u}{(1+x)} = (1+x)e^x$$

This is first order LDE of the type

$\frac{du}{dx} + P(u) = Q$. where

$$P = -\frac{1}{1+x}, \quad Q = (1+x)e^x$$

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$$IF = e^{\int\left(-\frac{1}{1+x}\right)} = e^{-\log(1+x)} = e^{\log(1+x)^{-1}}$$

$$IF = (1+x)^{-1} \qquad \boxed{IF = \frac{1}{1+x}}$$

$\Rightarrow$ Sol^n for first order LDE,

$$I.u = \int I\, Q\, dx + C$$

$$\Rightarrow \frac{1}{1+x} \cdot u = \int \cancel{(1+x)}\, e^x \frac{1}{\cancel{(1+x)}}\, dx$$

$$\Rightarrow \frac{1}{1+x} \times \sin y = \int e^x$$

$$\Rightarrow \boxed{\sin y = (1+x)\, e^x}$$

$$\boxed{\frac{\sin y}{1+x} = e^x + C}$$

Q19) Solve $\cos y - x \sin y \frac{dy}{dx} = \sec^2 x$.

$\Rightarrow$ Given $\cos y - x \sin y \frac{dy}{dx} = \sec^2 x$.

Divide by x.

$$(-\sin y)\frac{dy}{dx} + \frac{1}{x}\cos y = \frac{\sec^2 x}{x} \longrightarrow (I)$$

$\Rightarrow$ This DE of the form $f'(y)\frac{dy}{dx} + f(y)\, P = Q$.

where P & Q are the functions of x (or) const.

Here $f(y) = \cos y$ Put $\cos y = u$

$$\boxed{\cos y \rightarrow y \rightarrow x}$$

$$-\sin y \frac{dy}{dx} = \frac{du}{dx}$$

Eq^n (I) becomes

$$\frac{du}{dx} + \frac{1}{x} u = \frac{\sec^2 x}{x}$$

$\Rightarrow$ This is first order LDE of the type $\frac{du}{dx} + pu = Q$.

where $p = \frac{1}{x}$, $Q = \frac{\sec^2 x}{x}$

$$IF = e^{\int p\,dx} = e^{\int (\frac{1}{x})dx} = e^{\log x} = x.$$

$\Rightarrow$ Sol^n for first order LDE

$$u.I = \int I.Q\,dx + c$$

$$\Rightarrow u.x = \int x \frac{\sec^2 x}{x} dx + c$$

$$\Rightarrow u.x = \tan x + c$$

Back substitute $u = \cos y$

$$\boxed{\cos y.u = \tan x + c}$$

Q20) Solve $3y^2 \frac{dy}{dx} + 2xy^3 = 4xe^{-x^2}$

$\rightarrow$ Given $3y^2 \frac{dy}{dx} + 2xy^3 = 4xe^{-x^2}$

Divide by $3y^2$

$$\frac{dy}{dx} + \left(\frac{2x}{3}\right)y = \frac{4xe^{-x^2}y^{-2}}{3}$$

$\Rightarrow$ This is Bernoulli's D.E of the type

$\frac{dy}{dx} + Py = Qy^n$. with $n = -2$.

where $P = \frac{2x}{3}$ $\quad Q = \frac{4xe^{-x^2}}{3}$

Put $y^{1-n} = u$

$\Rightarrow y^3 = u$

diff wrt x.

$$3y^2 \frac{dy}{dx} = \frac{du}{dx} \Rightarrow \frac{dy}{dx} = \frac{y^{-2}}{3}\frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{1}{3y^2}\frac{du}{dx}$$

$\Rightarrow$ Eq^n becomes.

$$\frac{y^{-2}}{3}\frac{du}{dx} + \left(\frac{2x}{3}\right)y = \frac{4xe^{-x^2}}{3}$$

$$\frac{1}{3y^2}\frac{du}{dx} + \left(\frac{2x}{3}\right)y = \frac{4xe^{-x^2}}{3y^2}$$

multiply by $3y^2$.

$$\frac{du}{dx} + (2x)y^3 = 4xe^{-x^2}$$

but $y^3 = u$

$$\Rightarrow \frac{du}{dx} + (2x)u = 4xe^{-x^2}$$

$\Rightarrow$ This is first order LDE of form $\frac{du}{dx} + pu = Q$.

with $P = 2x$, $Q = 4xe^{-x^2}$.

$$IF : I = e^{\int P dx} = e^{\int 2x\, dx} = e^{2\left[\frac{x^2}{2}\right]} = e^{x^2}$$

$\Rightarrow$ Sol^n for first order LDE.

$$u.I = \int I Q dx + C$$

$$\Rightarrow u e^{x^2} = \int e^{x^2}(4xe^{-x^2})\, dx + C$$

$$\Rightarrow u.e^{x^2} = 4\int x\, dx + C$$

i.e $ue^{x^2} = 4\left[\frac{x^2}{2}\right] + C$

$$\Rightarrow ue^{x^2} = 2x^2 + C$$

back substitute $u = y^3$

$$\boxed{y^3 e^{x^2} = 2x^2 + C}$$ ✓

..... This is the END